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NEWS

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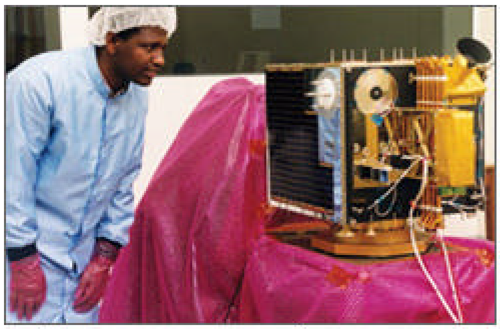
Countdown to FedSat launch for La Trobe space researchers

Navigation, timing and other technologies could become more accurate thanks to La Trobe University's participation in FedSat - Australia's first scientific satellite in three decades.



Making GPS systems more efficient: Mr Yizengaw tests in Melbourne before the launch later this year.

The FedSat 'micro-satellite' - a 50 cm cube weighing about 55 kg - will go into orbit from a rocket scheduled to be launched from Japan on 1 November 2002.

La Trobe University senior lecturer and space physicist, Dr Elizabeth Essex, is project leader for one of four major scientific studies using FedSat when it orbits from pole to pole 800 km above the earth. Her project will investigate applications of satellite communications, GPS technologies and the earth's magnetic field.

Originally developed for American defence, the GPS (Global Positioning System) is now widely used for navigation, - from berthing a passenger liner to finding a street in an on-board automobile location system - as well as for timing and other scientific purposes.

Working with a team that includes Doctoral student, Endawoke Yizengaw, and Master's student, Rudy Birsa, Dr Essex will investigate aspects of GPS signal distortion.

Signals transmitted from 30 GPS satellites orbiting at an altitude of 20,000 km have to pass through the ionosphere en route to earth where the density of electrons and other factors distort the signals. This affects the accuracy of the information used for navigation and other precision applications.

GPS satellite range signals are delayed according to the density of electrons in different parts of the ionosphere.

'It is vital that we devise means to correct these errors and we can only do this once we understand more about the ionisation distribution and its effect on the signals as they pass through the ionosphere,' Dr Essex says.

Her team is using the first FedSat project to look at different aspects of this problem. Mr Yizengaw, who graduated in physics from the University of Addis Ababa and as a Master of

Atmospheric Physics from Tromso University in Norway, will study the total electron content in the ionosphere.

His objective is to calculate the density of electrons and use this information to correct errors in the GPS signals.

Mr Birsa is researching scintillation - amplitude and phase variations - as a factor in signal irregularities that occur as GPS signals traverse the ionosphere. Pinpointing the volume of scintillation helps to understand irregularities in the low latitude ionosphere, he says.

FedSat will permit the recording of 'GPS slices' of the ionosphere, enabling a 3D moving picture to be constructed.

Dr Essex says FedSat will also help alleviate a problem with GPS satellite transmitting systems in the southern hemisphere. 'Because most of our hemisphere is water, there are fewer ground receiving stations. FedSat will act as a receiving station, thereby providing information on the Southern Hemisphere ionosphere.'

As well as scientific research on the ionosphere, FedSat's GPS receiver will allow accurate measurement of the satellite's position to help determine satellite orbits more precisely.

La Trobe's role in the FedSat project is as a member of the Cooperative Research Centre for Satellite Systems. The centre includes 12 Australian organisations and aims to develop domestic expertise in satellite technologies to help industry and the commercialisation of intellectual property.

FedSat has received an Australian Government AusIndustry grant of $2 million. Culminating five years of planning it will be launched by a National Space Development Agency of Japan rocket.

La Trobe's Department of Physics is a leader in space research and study. One of the few Australian universities that offers a Bachelor of Science (Space Science) degree, it also heavily involved in managing the Tasman International Geospace Environment Radar (TIGER), part of SuperDarn, a network of high frequency radars used to study the ionosphere.

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